ECE 447

Lesson 32
Random Processes



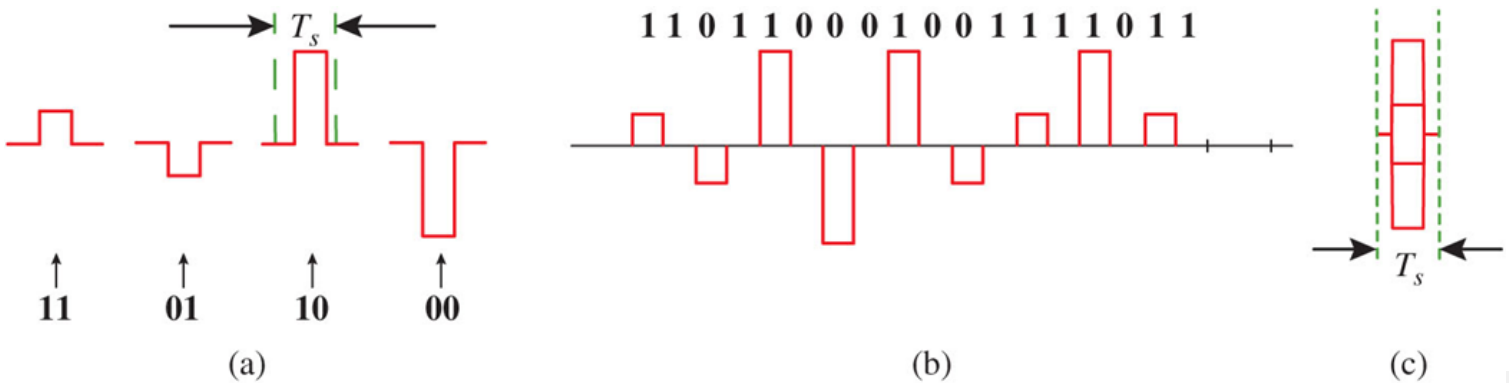
UNITED STATES
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ACADEMY

Schedule and Admin

- Schedule.
  - Lesson 33 - SDR Lab 7: ASCII messaging (Tuesday, 10 Nov)
  - Lesson 34 - **GR2 Review Day** (Friday, 13 Nov)
  - Lesson 35 - GR2 (Tuesday, 17 Nov) – Lessons 18-32, Chapters 5-8
  - Lessons 36-38 - Binary Digital System Performance, Error Correction
  - Lesson 39 - SDR Lab 8: Capstone image transfer
  - Lesson 40 - Advanced topics: OFDM, MIMO, CDMA
  - Lesson 41 - Course review
- Admin
  - **HW7**. Due tonight (Lesson 32, 6 Nov) to Gradescope.
  - **HW8**. Due Lesson 34 (Fri, 13 Nov) to Gradescope.
  - **Lab 7**. Assigned Lesson 33 (Tue, 10 Nov); due Lesson 37 (Mon, 23 Nov).

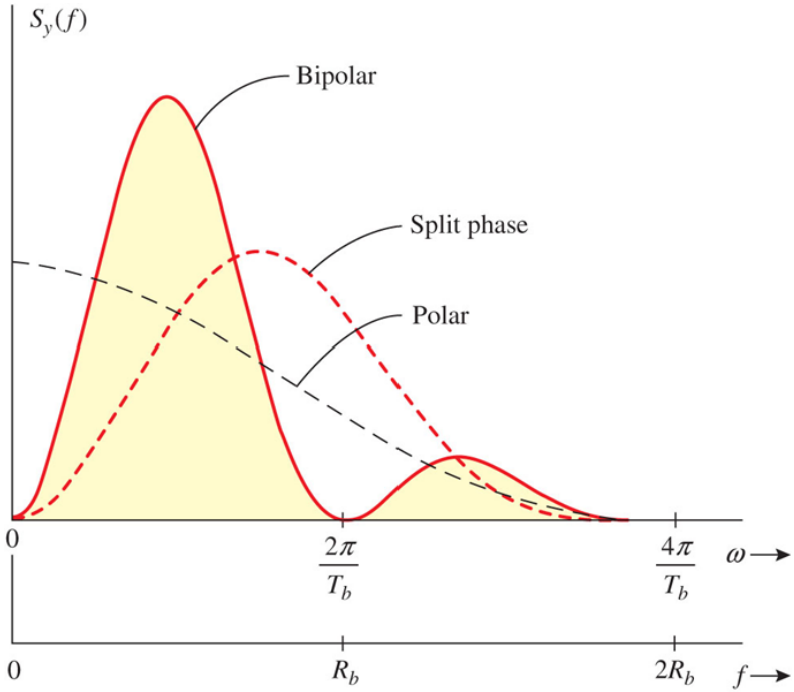
Review: M-ary Baseband Signaling

- Assign multiple bits to each pulse - to transmit n bits per pulse, you need $M = 2^n$ pulses (which is where the M in M-ary comes from)
- M-ary PAM: amplitudes define the pulses (other types of M-ary)
- Not a free lunch! See Example 6.6 in textbook - 4-ary PAM results in the same bandwidth as binary, but requires **5x** the original signal power
- New term! **Baud rate**: pulse or symbol rate. Ex: if using 16-ary PAM at a baud rate of 250 MBd (mega baud), the bit rate is 1 Gbps
- Baud rate focuses on physical transmission speed and should be used with Nyquist's theorem (2 bps/Hz $\rightarrow$ 2 baud/s/Hz)



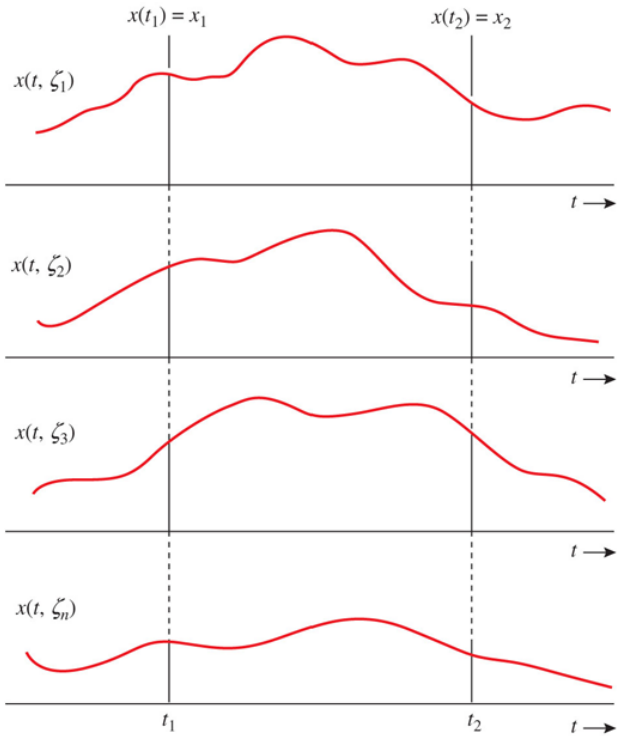
REVIEW

- Baseband signal bandwidth is inversely related to what?
- Rate of information transfer involves what?
- The figure on the right is for what type of line code?
- How does the bandwidth change for "full-width" pulses?



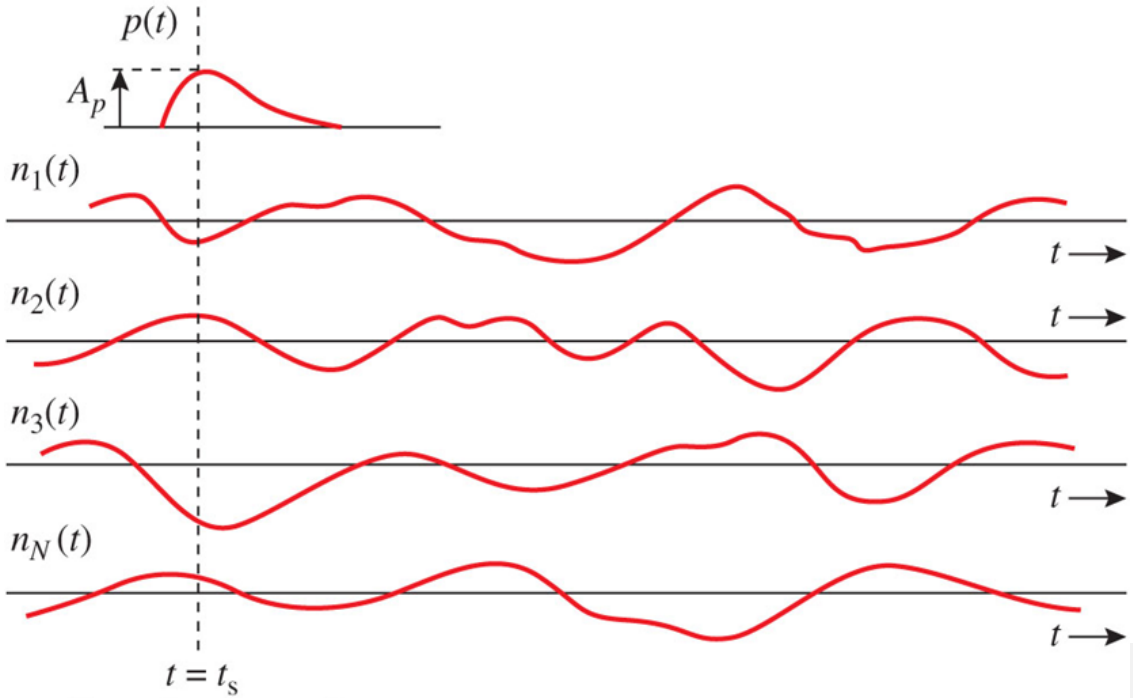
Random Variables to Random Processes

- Random process: a RV that is a function of time (aka stochastic process)
- RV: number of cars on a specified length of I-25 at a specific time every day
- RP: number of cars on a specified length of I-25 at every time throughout a day
- Instead of a collection of sample points mapped to real numbers (RV), we have a collection of sample *functions* that make up the entire sample space, i.e. a RP **ensemble**



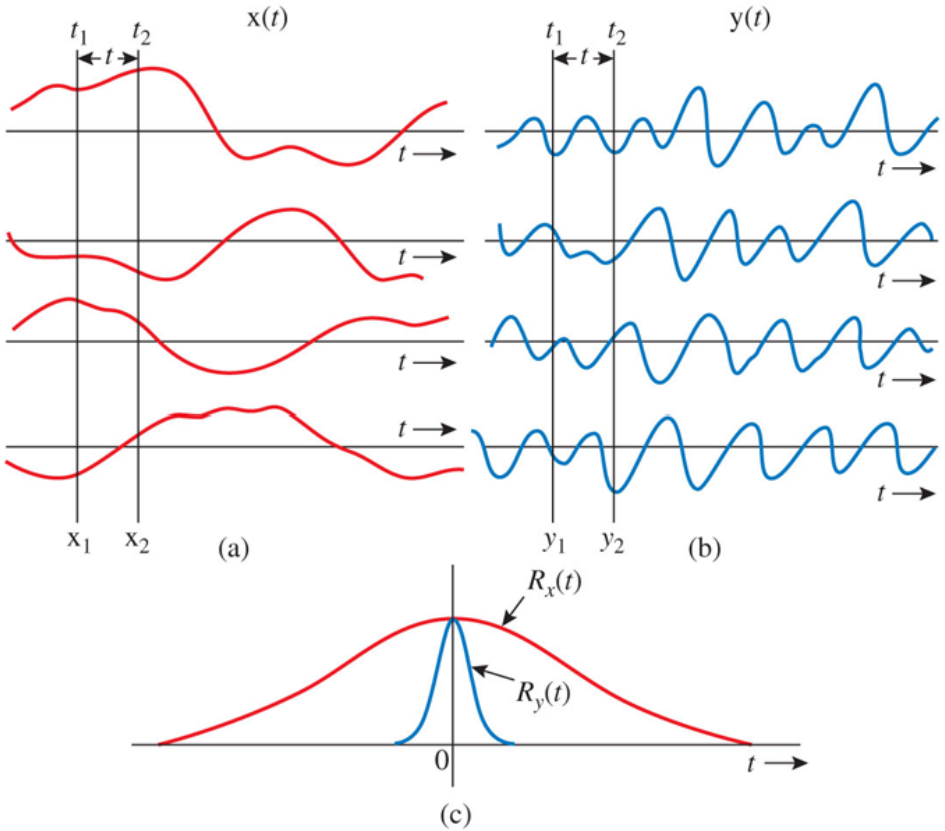
Describing RPs

- Can't always describe RP analytically - have to derive statistics from experimental data
- Complete info of multiple dependent RVs given by joint PDF
- Usually just want first- and second-order statistics
- **Ensemble statistics:** hold at a constant value of time and take statistics across RP ensemble at that time
- Example: threshold detection of polar signal



Spectral Content of RPs

- Autocorrelation again!
- $R_x(t_1, t_2) = \overline{x_1 x_2} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 x_2 p_x(x_1, x_2; t_x, t_2) dx_1 dx_2$
- Will you have to use this? Not explicitly in this class
- How do we get the PSD from the autocorrelation function?

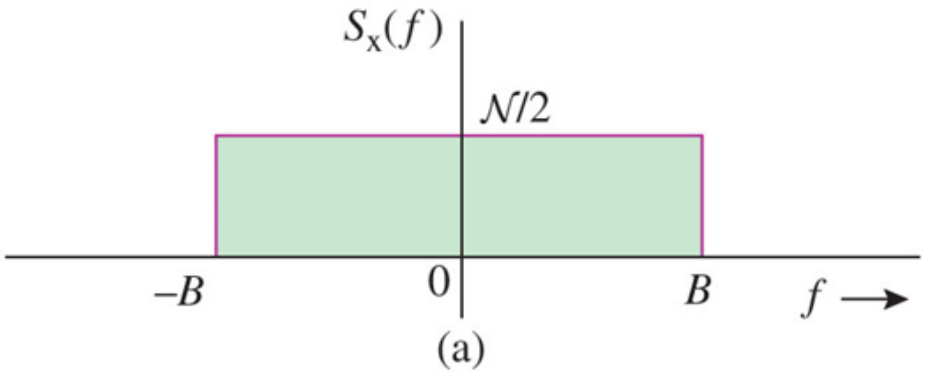


Categories of RPs

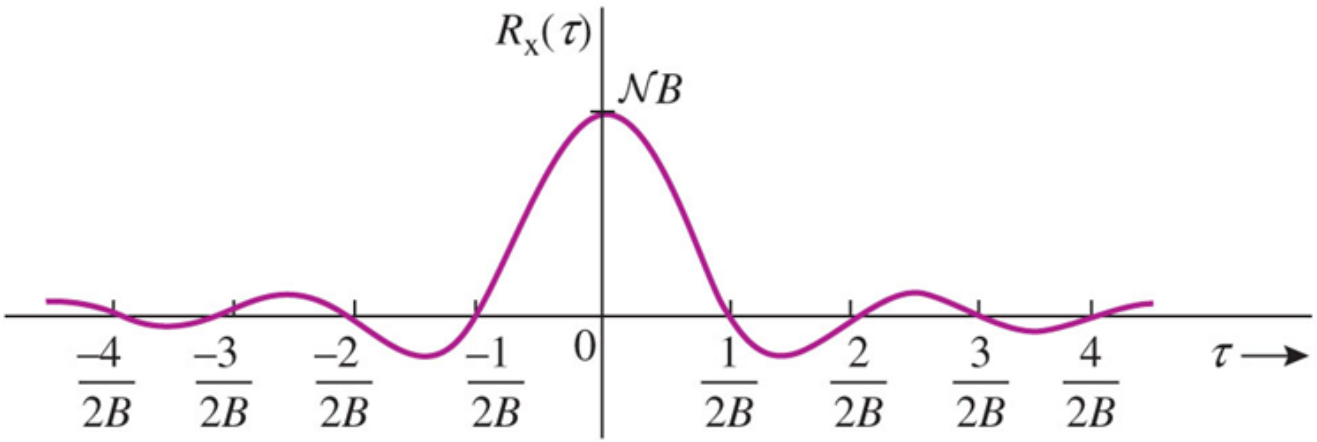
- **Stationarity**: a stationary RP's statistical characteristics do not change with time - $p_x(x;t) = p_x(x)$ and $R_x(t_1, t_2) = R_x(t_2 - t_1) = R_x(\tau)$
- Examples: channel noise is stationary; temperature is not
- **Wide-sense Stationarity**: doesn't meet the strict definition of stationary, but still has mean $\overline{x(t)}$ and autocorrelation $R_x(\tau)$ that are independent of time shifts
- Truly stationary RP do not exist; WSS RPs do exist
- **Ergodicity**: RPs whose ensemble averages (at constant times) are equivalent to the time averages (remember this from Chapter 3?) are WSS ergodic processes
- If process is ergodic, then we only need one sample function to find the ensemble averages

PSD

- Average power of a RP: $P_x = \overline{|x|^2} = R_x(0) = \int_{-\infty}^{\infty} S_x(f)df = 2\int_0^{\infty} S_x(f)df$



(a)



(b)